

CT Imaging Turnaround Time & Radiology Workflow Optimization Project Proposal

Business Problem

A radiology department has been noticing that CT turnaround times are inconsistent. Some examinations are completed and reported very quickly, while others take significantly longer. The challenge is that without a detailed breakdown of the entire workflow, it is difficult to determine where these differences are coming from. Is the delay happening at scheduling, during the scan itself, or in the reporting phase? This lack of visibility makes it hard to improve performance or set realistic expectations for referring departments and patients.

Project Objectives

- i. Break down the entire CT process from order to final report, measuring the time spent at each stage.
- ii. Determine exactly which stage (scheduling, scanning, or reporting) is contributing the most to overall turnaround time.
- iii. Understand how variables like exam type, priority level, shift, scanner, and radiologist affect processing times.
- iv. Offer specific, data-backed suggestions for workflow improvements, resource allocation, and staffing adjustments.

Datasets

For this analysis, we will use a synthetic dataset created to realistically model CT imaging operations. It will contain approximately 20,000 CT examinations, providing a robust sample for meaningful insights. The dataset will include crucial fields such as:

Timing Data: Timestamps for Order, Scheduled Time, Arrival, Scan Start, Scan End, Report Start, and Report Finalized.

Exam Details: Exam type, body part scanned, priority level, and whether contrast was used.

Operational Context: The radiologist and technologist assigned, scanner ID, shift time, patient category (Inpatient, Outpatient, or ED), and whether a repeat scan was required.

This data will allow us to analyze each step of the workflow independently and identify where improvements can have the greatest impact.

Business Questions

1. What is the average and median CT turnaround time from order to final report? What is the 90th percentile?
2. Which step creates the greatest delay? Is it scheduling, scan duration, or radiologist reporting time?
3. Which CT examination types have the longest turnaround times?
4. Which shift experiences the greatest delays? Does priority level influence how quickly an exam is completed?
5. Which scanner has the highest workload? Which radiologist has the fastest reporting times?
6. Are ED CT examinations processed faster than inpatient or outpatient exams?
7. What percentage of exams exceed the SLA target, and where do those delays typically occur?

Project Workflows

- ✓ We will follow a clear, step-by-step process to ensure the project is organized and delivers value:
- ✓ Project Kick-off and Data Review: We will meet to finalize the project scope and discuss the data, KPIs and metrics to use.
- ✓ Data Processing and Analysis (SQL and Excel): We will use SQL to query the data and calculate all necessary KPIs, including average TAT, median TAT, and stage-by-stage breakdowns. Excel will be used for validation and supplementary analysis.
- ✓ Dashboard Development (Power BI or Tableau): We will build an interactive dashboard split into three clear pages:

- A high-level snapshot with KPI cards for total exams, average TAT, SLA compliance, and repeat rate, plus monthly trends and volume by department.
- A visual representation of the complete CT journey from order to report, showing the average duration between each stage.
- Detailed breakdowns of TAT by scanner, radiologist, shift, exam type, priority, and patient category to pinpoint where delays are concentrated.
- ✓ We will prepare a final report summarizing our findings and present it, highlighting the biggest bottlenecks and our data-backed recommendations for improvement.